

Project 6: Mean–variance modelling and Value-at-Risk

ARMA + GARCH on a financial returns series

Aymen Ben Brik
aymen.benbrik@esprit.tn
Esprit School of Business

2025–2026

Context

This project closes the module by combining Chapters 3–6 in a single volatility study. Starting from a real (or pre-supplied) financial returns series, you will:

1. fit a mean model (ARMA) on the level;
2. test for ARCH effects on the residuals;
3. fit a GARCH (or extended) model on the conditional variance;
4. validate the joint specification through standardised residuals;
5. produce a one-day Value-at-Risk and back-test it.

The deliverable is a self-contained risk-management report.

Deliverables

- Reproducible `project6_<name>.py` with seed 42.
- A returns series of length ≥ 1500 (daily). You may use the supplied `data/financial_returns.csv` or bring your own (clear source citation).
- Report PDF (10–12 pages) including all figures, all numerical outputs, the back-test of the VaR.
- Optional 12-min oral defence.

Exercise 1 – Descriptive statistics and stylised facts

- (1) Plot the price series P_t and the log-returns $r_t = \log(P_t/P_{t-1})$.
- (2) Report descriptive statistics on r_t : mean, standard deviation, skewness, excess kurtosis. Compare with what would be expected under a Gaussian model.
- (3) Plot the sample ACF of r_t and of r_t^2 up to lag 50. Confirm the two stylised facts: returns are nearly uncorrelated, but their squares are strongly autocorrelated.

Exercise 2 – Mean model: ARMA

- (1) On r_t , run ADF and KPSS to confirm stationarity.
- (2) Identify candidate ARMA orders from ACF/PACF and fit three candidates by MLE. Report AIC, BIC, Ljung–Box on residuals.
- (3) Choose the best mean model. If no significant ARMA structure is present, justify keeping a constant-mean model.

Exercise 3 – Detecting ARCH effects

- (1) On the residuals $\hat{\varepsilon}_t$ of the chosen mean model, run Engle's LM test at lags 5, 10, 20. Report the table.
- (2) Plot the ACF of $\hat{\varepsilon}_t^2$ and confirm the visual evidence.
- (3) Conclude: GARCH modelling is needed.

Exercise 4 – Variance model: GARCH(1, 1)

- (1) Fit the joint mean–variance model $r_t = \mu_t + \varepsilon_t$, $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ with Gaussian innovations. Report all estimates with standard errors.
- (2) Verify the stationarity condition $\hat{\alpha}_1 + \hat{\beta}_1 < 1$ and report the implied unconditional variance.
- (3) Refit with a Student- t innovation distribution and compare AIC, BIC. Which one fits better? What does this say about the tails of the data?

Exercise 5 – Diagnostics on standardised residuals

- (1) Compute $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ from the chosen GARCH.
- (2) Plot the ACF of $\hat{z}_t$ and $\hat{z}_t^2$ up to lag 20. Run Engle's LM test on $\hat{z}_t^2$.
- (3) Plot a QQ-plot of $\hat{z}_t$ vs. the assumed distribution. Comment on the marginal fit.

Exercise 6 – Volatility forecast and Value-at-Risk

- (1) On the chosen joint model, plot the in-sample $\hat{\sigma}_t$ together with the time series r_t . Identify visually the most turbulent periods.
- (2) Compute the multi-step variance forecast $\hat{\sigma}_{T+h}^2$ for $h = 1, \dots, 30$. Plot it together with the unconditional variance.
- (3) Compute the one-day Value-at-Risk at $\alpha \in \{1\%, 5\%\}$, both conditionally ($\hat{\sigma}_{T+1}$) and unconditionally ($\hat{\sigma}$).

Exercise 7 – Bonus: VaR back-test

On a hold-out test set (last 250 business days), compute the one-day rolling conditional VaR at $\alpha = 1\%$ and $\alpha = 5\%$. Count the number of *exceedances* (days where the realised loss exceeded the predicted VaR).

- Expected number of exceedances at level α over N days: αN .
- Run a Kupiec test (`statsmodels` or by hand) to check whether the realised exceedance count is consistent with the nominal α .

Marking grid (out of 100)

Criterion	Points
Descriptive statistics + stylised facts	10
ARMA mean model + Box–Jenkins	15
Engle LM + diagnosis of ARCH effect	10
GARCH(1, 1) Gaussian + Student- t comparison	20
Diagnostics on standardised residuals	15
Forecast + Value-at-Risk	15
Bonus: VaR back-test (Kupiec)	10
Reproducibility + report quality	5
Total	100

Aymen Ben Brik
aymen.benbrik@esprit.tn